

System Design Basics – Refined Notes

1. Latency

Time taken to process a request and get a response.

Latency = Network Delay + Computational Delay

2. Throughput

Amount of work processed per unit time (requests/sec or bps).

3. Availability

System is operational when needed.

Fault Tolerance $\propto$ Availability

4. Replication vs Redundancy

Replication: Copy of data across nodes.

Redundancy: Duplicate systems for failover.

5. Consistency

All clients see same data.

Types: Strong, Eventual, Weak

6. CAP Theorem

Choose 2: Consistency, Availability, Partition Tolerance

7. Logical Clock

Orders events in distributed systems.

8. Load Balancing

Distributes traffic across servers.

9. Caching

Stores frequent data for faster access.

Eviction: LRU, MRU, LFU, FIFO, LIFO

10. RDBMS

Pros: Integrity, ACID

Cons: Rigid schema, scaling issues

11. NoSQL

Key-Value (Redis), Document (MongoDB), Column (Cassandra), Graph

12. Polyglot Persistence

Using multiple DBs together.

13. Normalization vs Denormalization

Normalization reduces redundancy.

Denormalization improves performance.

14. Indexing

Speeds up reads but slows writes.

15. Communication

Sync (blocking) vs Async (non-blocking)

16. Web Server

Handles HTTP requests.

17. Protocols

Push, Pull, Long Polling, WebSocket, SSE